Predictive Analytics for Sales Forecasting

Submitted in partial fulfillment of the requirements for the award of the degree of

Master of Science in **Data Science**

Submitted By:

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Session: 2024 -2026

Under the guidance of

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CERTIFICATE

Certified that this project report titled “Predictive Analytics for Sales Forecasting” is the bonafide work of "Piyanshu Singh" and REG.NO "O24MSD110133” who carried out the project work under my supervision.

As far as I know, the material presented in this thesis has not been submitted for any other degree or diploma.

Supervisor Signature:

____

Vikas Kumar Atray

Designation:

Date: May 17, 2026

Place: Chandigarh University

DECLARATION

Hereby, I Priyanshu Singh(UID: O24MSD110133) declare that the project report entitled "PREDICTIVE ANALYTICS FOR SALES FORECASTING IN FINANCE SECTOR" submitted to Chandigarh University in partial fulfilment of the requirement of the degree of Master of Science in Data Science is a record of original work done by me under the supervision of Vikas Kumar Atray.

I also declare that the work reported in the project has not been nor will be submitted by me either in whole or in part to any other degree or diploma in this institution or any other institution

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ABSTRACT

This project is a comprehensive implementation of predictive analytics solutions for sales forecasting for the financial and insurance verticals. This study is called Predictive Analytics for Sales Forecasting in the Finance Sector, and the data set it uses includes historical information from insurance agencies from 2005 to 2014 (213,328 records from 1,623 insurance agencies in 6 states of the United States).

The goal of this project is to build and test predictive models which would be able to predict the written premium accurately from past financial metrics such as loss ratio, retention ratio, new business premium, and incurred losses. The methods that were employed are Linear Regression (baseline), Random Forest Regressor, and Facebook Prophet for time-series forecasting, all three of which were implemented and compared.

This project uses Python as the main programming language, and some of the libraries utilized are Pandas, NumPy, Scikit-learn, and Prophet for data processing and machine learning respectively. Structured data was queried using SQL (SQLite) whereas Power BI was used to create an interactive 3-page dashboard, offering a descriptive and predictive analytical insight.

The best results were obtained by the Random Forest Regressor on log-transformed premium values with an R^2 score of 0.848, a Mean Absolute Error (MAE) of 0.505 and an RMSE of 0.858. The two best indicators of total premium volume are incurring losses and new business written premium, according to key findings. Commercial lines grew 114% over the last decade, and Ohio made up 58% of the total written premium portfolio.

The project builds the existing knowledge in the field of predictive analytics in finance and offers valuable information for data driven recommendations to enhance the accuracy of sales forecasting in the insurance industry.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ML	Machine Learning
EDA	Exploratory Data Analysis
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
R^2	R-Squared (Coefficient of Determination)
RF	Random Forest
LR	Linear Regression
CL	Commercial Lines
PL	Personal Lines
SQL	Structured Query Language
BI	Business Intelligence
DFD	Data Flow Diagram
UML	Unified Modelling Language
ER	Entity Relationship
API	Application Programming Interface
CSV	Comma-Separated Values
KPI	Key Performance Indicator
YoY	Year-over-Year
UI	User Interface

CHAPTER 1: INTRODUCTION

1.1 Background of the Project

The insurance and financial services sector create huge transactional data on a daily basis. Insurance agencies monitor written premiums, policy renewals, claims, loss ratios and new business, as well as thousands of client accounts. Despite the wealth of historical information available, a lot of financial organizations are still using manual approaches, gut instinct or cruder statistical models to forecast sales success in the future. Predictive Analytics is the predictive modelling process to identify future events from past trends using statistical algorithms, machine learning models, and data mining techniques. As it relates to insurance sales, predictive analytics can revolutionize the way that agencies allocate resources, target their sales goals and find markets where growth is going to occur before it does. Being able to predict written premium volumes one to three years ahead gives insurance agencies an edge in the competitive arena. The financial services industry - including banking, insurance, investment management and financial advisory - is one of the most data-intensive sectors in the global economy. Recent studies indicate that companies with sophisticated forecasting methods using advanced analytics can outperform companies with traditional forecasting methods in terms of forecast accuracy and operational efficiency. Predictive analytics is now possible for every organization, thanks to the rise of cloud computing, open-source machine learning libraries, and business intelligence platforms. This project is targeted specifically at the insurance sub-sector of the finance sector with a real-world dataset of agency performance measures covering 11 years (2005 - 2015) of data. The data includes 1,623 independent insurance agencies in six states in the United States, offering a variety of commercial lines (CL) and personal lines (PL) insurance products. The project investigates how to use several machine learning approaches to predict written premium sales and compare the models' performances using conventional statistical measures.

1.2 Problem Statement

One of the major challenges for insurance agencies and financial institutions in the regional insurance market is the lack of ability to correctly estimate future volumes of written premiums from historical volumes. Traditional forecasting approaches focus primarily on year-over-year growth projections and/or a manual examination of the past, without taking into consideration the various factors that influence financial indicators like loss ratios, retention rates, acquisition of new business, market specific factors, etc. The lack of a systematic, data-driven forecasting framework means there are several operational issues. Agencies cannot allocate their resources and establish realistic sales goals without first being able to do the above. Second, without knowledge of expected claim trends and volumes, underwriters are unable to sufficiently price the risk. Third, management does not have visibility of what product lines, geographic areas and agency segments are likely to be more successful or less successful in the next several years. To meet these challenges, this project builds and tests predictive models that will be able to estimate insurance sales (written premium) based on financial performance indicators from past years and compare their performance. The proposed system is designed to fill the gap between the amount of available data and the ability to take action to achieve the desired forecasting.

1.3 Objectives of the Project

To conduct a comprehensive literature review and understand the significance of predictive analytics in sales forecasting in the finance industry. Analyses past sales data and financial metrics to create predictive models for insurance premium forecasting. Compare the accuracy and effectiveness of multiple predictive models such as Linear Regression model, Random Forest Regressor model and Facebook Prophet model using standard evaluation metrics (MAE, RMSE, R^2). To learn about various predictive analytics methods to forecast sales and to understand the best model for the insurance industry. To make possible recommendations to enhance sales forecasting through predictive analytics in the finance and insurance sectors.

1.4 Scope of the Project

This project's scope includes the entire data pipeline from data acquisition to business recommendations. The project includes the following activities:

- A project involving Exploratory Data Analysis (EDA) for the analysis of 213,328 records of insurance agency performance over the time period 2005-2014.
- Data preprocessing (sentinel values of treatment, feature engineering, log transformation of the target variable, etc.), and train test splitting. Design and testing machine learning sales forecasting models.
- Utilization of Structured Query Language (SQL) for aggregation of agency performance and financial health assessment using the SQLite language.
- Creating an Interactive Dashboard with Microsoft Power BI with 3 Analytical Pages. Structured academic report including documentation of findings, methodology and recommendations.

This project is restricted to the insurance industry and uses the publicly available Kaggle data. It does not involve real-time data streaming, proprietary insurance systems integration, or deployment in a production environment.

1.5 Existing System Overview

The three main types of sales forecasting currently used in insurance and finance are, there are three major methods of current sales forecasting used in insurance and finance: The first are the manual projections, in which sales managers estimate future performance based upon past averages and growth rate assumptions. They are all subjective and open to human bias and can be prone to inaccurate forecasts which fail to take into account underlying market dynamics. The other types of simple models are basic statistical models like moving averages and simple linear regression. These models are more objective but cannot incorporate non-linearities and complex interactions among financial variables. They also have to deal with the variable and skew nature of insurance premiums. The third category includes commercial enterprise resource planning (ERP) systems and actuarial software with a certain forecasting component, but are costly, specialized software, and difficult to modify exploratory analytical workflows. These systems tend to be opaque and offer no detail at the feature level of what is driving sales. Current systems have a few drawbacks: They have no machine learning ability, they don't provide quantitative feature importance, they don't provide any time-series aware forecasting, and they don't allow for predictive insights to be visually communicated to business users.

1.6 Proposed System Overview

The suggested system is an insurance sales forecasting application that utilizes end-to-end predictive analytics pipeline. The system takes historical performance data of agency as input and outputs forecast results with a series of well-defined analytical phases.

It starts with data ingestion and exploratory data analysis, followed by systematic pre-processing to tackle data quality challenges such as sentinel value treatment, feature engineering, and target variable normalization. Three predictive models are then trained and evaluated: Linear Regression model as a baseline model, a Random Forest Regressor model to capture the non-linear patterns, and a Facebook Prophet model to forecast the time-series specific patterns.

The system outputs performance metrics of the models (MAE, RMSE, R2), feature importance rankings, and comparison charts of forecast vs actual. These are fed into a three-page Power BI dashboard that gives stakeholders descriptive, financial health and predictive analytics views. An SQL layer allows for structured queries to be used for aggregate reporting.

The proposed system is more advantageous than the current ones, as it integrates the transparency of machine learning (feature importance), the time-series aspect (Prophet), and the interactive visualization (Power BI) in a single analytical system, which is reproducible, developed entirely using open source tools.

1.7 Technologies Used

Technology	Version/Tool	Purpose
Python	3.13	Data analysis, ML modeling, preprocessing
Pandas	2.x	Data manipulation and analysis
NumPy	1.x	Numerical computation
Scikit-learn	1.x	Linear Regression, Random Forest, metrics
Facebook Prophet	1.x	Time-series forecasting
Matplotlib / Seaborn	3.x	Data visualization
SQLite (Python)	Built-in	Database creation and SQL queries
Microsoft Power BI	Desktop	Interactive dashboard
Jupyter Notebook	VS Code	Development environment
Git / GitHub	Latest	Version control and portfolio hosting
Kaggle	Platform	Dataset source

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction to Predictive Analytics in Finance

Predictive analytics is one of the advanced analytics subfields, which involves forecasting future events or outcomes based on historical data, statistical algorithms, and machine learning techniques. Predictive analytics has proved to be a key ability in the finance industry, where businesses are looking to make data-informed decisions to gain a competitive edge. The financial services sector spans banking, insurance, investment management, and lending and creates huge amounts of transactional and behavioural data that can prove to be very insightful when analysed correctly and in good time, allowing for predictions to be made.

In recent years, the use of predictive analytics in financial sales forecasting has grown significantly, as financial institutions have realized that conventional methods of forecasting are not effective in today's complex market conditions, and as more data has become available in the historical record, open source machine learning frameworks have proliferated. Machine learning-based forecasting has been shown to consistently outperform traditional statistical forecasting techniques in terms of forecast error.

Predictive Analytics is used in insurance for several purposes: premium pricing, predicting claims, forecasting customer retention and managing the pipeline of new business. Being able to predict written premium volume at the agency level is especially useful for regional insurance groups that have multiple lines of business and multiple independent agents in various states.

2.2 Review of Related Studies

Study 1: Prediction for Insurance Premiums Based on Random Forest and Multiple Linear Regression

A study published on researchgate (2023) compared the performance of the Random Forest model to the Multiple Linear Regression model in predicting insurance benefits and premium amounts. The researchers tested both models on an actual insurance dataset and demonstrated the superiority of the Random Forest model over Linear Regression in prediction accuracy from the dataset. The study also suggested that although the accuracy of Random Forest was higher, it was not easily interpretable as was Linear Regression where the relationship between the input variables and the premium outcomes can be easily explained.

The authors found that both models are useful for their respective organizational purposes: Random Forest is better for the accuracy of the task, whereas Linear Regression is suitable for people who want to know more about forecasting and need to understand the process. The approach taken in this project to implement both models and evaluate the trade-offs fits well with this finding. The same trend has been observed in our results, which shows that the R^2 value of the Random Forest is higher than the Linear Regression's on the insurance agency dataset, with $R^2 = 0.848$ and $R^2 = 0.309$, respectively.

Study 2: Random Forest Regression with Hyper Parameter Tuning for Medical Insurance Premium Prediction

Prakash et al. (2022) conducted a research paper that showed the effectiveness of Random Forest Regression (with hyperparameter tuning) for medical insurance premium prediction in the International Journal of Health Sciences. The study was applied to a healthcare insurance dataset, showing that the model could be optimized with good hyperparameter values, which would lead to better performance. Data preprocessing, specifically feature selection, as well as outlier detection, was found to be essential for obtaining accurate prediction outcomes.

There are two major aspects of the paper that are relevant to this project. First, it confirms the selection of the Random Forest model as the main predictive model to forecast insurance premiums. Secondly it highlights the significance of the preprocessing pipeline used in this project, specifically the treatment of the heavily skewed target variable and the sentinel values (99999 placeholders).

Study 3: A Study of Impact and Applications of Predictive Analytics in Sales Forecasting

Kumar (2023) conducted an extensive review of the International Journal of Research in Applied Science and Engineering Technology on the various applications and applications of predictive analytics in various sectors including finance, retail, and healthcare. The study revealed that AI-driven predictive systems can crunch through vast amounts of historical and pipeline data and deliver much more accurate sales forecasts than human analysis. The organizations that used predictive sales forecasting saw an increase in revenue predictability, in efficient use of resources, and capability for better strategic planning.

A crucial aspect of this paper for this project is the fact that it provides the theoretical foundation to use machine learning for sales forecasting in the finance industry. The results validate the project's main thesis of generating reliable forward-looking premium forecasts through the analysis of historical agency performance with the right ML models. The paper also states that data quality and feature engineering are two challenges that can become a success factor, and therefore were taken into account in this project when making choices among the different pre-processing options.

Study 4: AI Driven Predictive Analytics for Financial Forecasting.

On the other hand, the IEEE conference paper (2024) explores the application of AI predictive analytics for financial forecasting in risk management and business planning. The study evaluated the performance of AI technologies for predicting financial results and their potential for real-time reporting in financial institutions. The study revealed that companies that adopted AI-driven forecasting systems outperformed those employing traditional statistical methods when it came to the accuracy of their financial performance forecasts.

The paper's main finding is that predictive analytics can be a game-changer in financial risk management. It determined that predictive models with multiple financial indicators outperformed models with single variable forecasting, which directly supports the multi-feature modeling approach taken for this project. The features used in our model (loss ratio, retention ratio, new business premium) are consistent with the multi-indicator strategy suggested by this research.

Study 5: Predictive Analytics in Financial Management: Enhancing Decision-Making and Risk Management

A systematic review of the use of predictive analytics to improve decision-making and risk management in financial services organisations was published on ResearchGate (2024). The review compiled results from diverse financial management studies and concluded that predictive analytics in financial management drive tangible benefits in terms of enhanced predictive accuracy, risk measurement, and operational efficiency. Specifically, the review referenced a survey conducted by the World Economic Forum (2023) which revealed that 54% of the financial sector firms faced challenges in recruiting individuals with the appropriate skills in analytics.

The literature review revealed a clear gap in the literature with regards to using predictive analytics in insurance sales forecasting at the agency level over multiple years—although there is a lot of literature in the e-commerce and banking sectors to support the use of predictive analytics in cost prediction for medical insurance and banking, respectively, there is limited published research that focuses on multi-year agency-level insurance sales forecasting using ensemble methods. This gap directly underlies the direction of the research of this project.

2.3 Comparative Analysis of Predictive Techniques

Technique	Accuracy	Interpretability	Best Use Case
Linear Regression	Low-Medium	High	Baseline, transparent models
Random Forest	High	Medium	Complex non-linear patterns
Facebook Prophet	Medium	High	Seasonal time-series
XGBoost	Very High	Low	Tabular data competitions
LSTM/Deep Learning	High	Very Low	Large sequential datasets

Considering the nature of the insurance agency performance data and the comparative analysis, the Random Forest Regressor model was chosen as the primary model because of its ability to provide a good balance of accuracy and interpretability. The interpretable baseline was Linear Regression and Facebook Prophet was added because of its time-series awareness.

2.4 Software Development Methodology

The project is undertaken in accordance with the methodology of Cross-Industry Standard Process for Data Mining (CRISP-DM) which is an industry standard method for data science and data analytics projects. The CRISP-DM is made up of six phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, Deployment.

The CRISP-DM methodology was chosen over the traditional software development models (Waterfall, Agile) as the methodology is specifically designed for analytical and machine learning projects that are primarily looking to deliver a predictive model which is different from a software application. It makes it possible to go back and refine the data preparation and modelling steps as a result of the evaluations performed, as necessary in this project when dealing with data quality concerns, e.g. sentinel values or distribution skewness.

2.5 Relevant Frameworks and Libraries

The scikit-learn library is an open-source Python machine learning library that includes implementations of Linear Regression, Random Forest Regressor, and evaluation metrics such as MAE, RMSE, and R^2. It was chosen because of its API's well documented use, reproducibility, and adoption in the data science community.

Facebook Prophet is a time-series forecasting open-source library, created by the Core Data Science team at Facebook. This method is designed to process datasets that have strong seasonal characteristics and multiple seasonalities, and has a good performance on the time-series with missing data and outliers. The additive model structure of Prophet is especially well suited for forecasting annual insurance premiums.

Microsoft Power BI is a business analytics service that offers interactive data visualization and business intelligence. It was chosen for dashboard development for a number of reasons including built-in support for DAX calculated columns, slicer-based filtering and multi-page report layouts.

SQLite is a small, embedded, serverless, relational database engine that comes with Python. It was employed to develop a structured database from the cleaned data and to run the SQL aggregation queries to analyze agency performance.

2.6 Research Gap

A review of literature clearly indicates that this project fills a gap in the literature. Although there is a considerable amount of research on predictive analytics for fraud detection in banking, predictive credit risk assessment and medical insurance cost prediction, there is limited published research focusing on multi-year agency-level insurance sales forecasting that integrates ensemble machine learning techniques with time-series forecasting and interactive business intelligence visualization within a single analytical framework.
In particular, some previous works focus on the performance of a single model only, or on forecasting in one domain only. The existing studies reviewed do not integrate the three models - Random Forest, Linear Regression, and Prophet - together and compare their

performance on insurance agency performance data, using explicit financial metrics like loss ratio, retention ratio, or new business premium as predictive features.

To fill this gap, this project will show a comparative performance study of three forecasting techniques on a real-world insurance dataset and discuss the results in a multi-page Power BI dashboard - all in one end-to-end reproducible, open-source analytical pipeline.

CHAPTER 3: SYSTEM ANALYSIS

3.1 Functional Requirements

FR No.	Requirement	Description
FR-01	Data Ingestion	The system shall load CSV data into Python DataFrame and SQLite database.
FR-02	Data Exploration	The system shall generate summary statistics and EDA visualizations.
FR-03	Data Preprocessing	The system shall handle sentinel values, encode categoricals, and apply log transformation.
FR-04	Model Training	The system shall train Linear Regression, Random Forest, and Prophet models.
FR-05	Model Evaluation	The system shall compute MAE, RMSE, and R^2 for each model.
FR-06	Visualization	The system shall generate forecast vs. actual charts and feature importance plots.
FR-07	SQL Querying	System shall execute aggregation queries on the SQLite database.
FR-08	Dashboard	The system shall display results in an interactive Power BI dashboard.

3.2 Non-Functional Requirements

NFR No.	Category	Requirement
NFR-01	Performance	Model training shall complete within 5 minutes on standard hardware.
NFR-02	Reliability	Preprocessing pipeline shall handle missing/sentinel values without data loss.
NFR-03	Reproducibility	All notebooks shall use fixed random seeds (random_state=42).
NFR-04	Scalability	The system shall handle datasets up to 500,000 records.
NFR-05	Usability	Dashboard shall be navigable by non-technical stakeholders.
NFR-06	Portability	The project shall run on Windows, Mac, and Linux environments.

3.3 Feasibility Study

Technical Feasibility

The project can be technically implemented with easily available open source tools. The data processing and model development are all covered by Python 3.13 and its libraries Pandas, Scikit-learn and Prophet. Power BI Desktop is a free download from Microsoft. SQLite comes built into Python and doesn't need to be installed. The dataset (213,328 records, 49 columns) is not too complex to process with the laptop hardware.

Economic Feasibility

There are low cost implications of the project. The software tools used (Python, Scikit-learn, Prophet, SQLite, Jupyter Notebook, GitHub) are all open source and free to use. Power BI Desktop is available for free for individuals and for educational purposes.

This data was obtained free of charge from Kaggle. No cloud computing or infrastructure costs since all processing is done locally.

Operational Feasibility

The project is within the scope of the academic year. The CRISP-DM methodology is structured and can be done in a 14-day sprint. This report, along with the other outputs such as Jupyter Notebooks, SQL query, and the dashboard (Power BI), are well-documented and can be maintained and extended by future researchers.

3.4 System Architecture

The system is architected as four layers: Data Layer, Processing Layer, Analytics Layer and Presentation Layer.

The Data Layer is the raw CSV data that is located in the data/raw/ directory as well as the processed data in the data/processed/ directory, and the SQLite database (insurance.db). The Processing Layer has three jupyter notebooks: 01_data_exploration.ipynb for data exploration, 02_preprocessing_modeling.ipynb for data preprocessing and modeling, and 03_sql_queries.ipynb for SQL-based analysis. The Analytics Layer is the trained models (Linear Regression, Random Forest, Prophet) and their prediction output files. The Power BI dashboard with its three pages and the project report are part of the Presentation Layer.

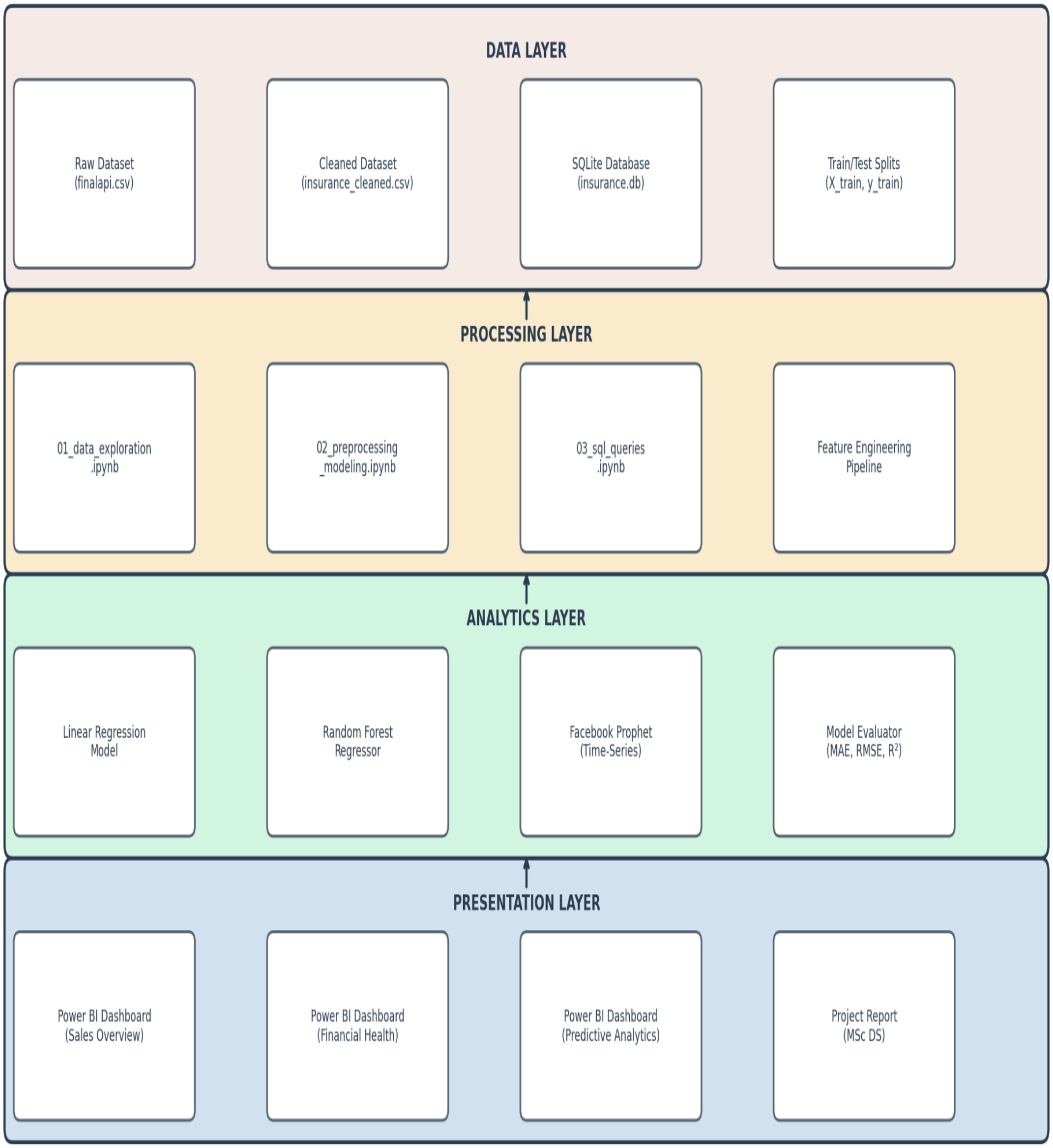


Figure 3.1: System Architecture Diagram

3.5 Data Flow Diagrams

DFD Level 0 (Context Diagram)

The Level 0 DFD displays the system as one process with the external entities and data flows. External users are: Data Analyst (user), Kaggle Dataset (data source), and

Business Stakeholder (report consumer). The key data flows include: Raw CSV data coming in from Kaggle to the system; Processed forecasts and insights from the system to the Data Analyst; and Dashboard visualizations to the Business Stakeholder.

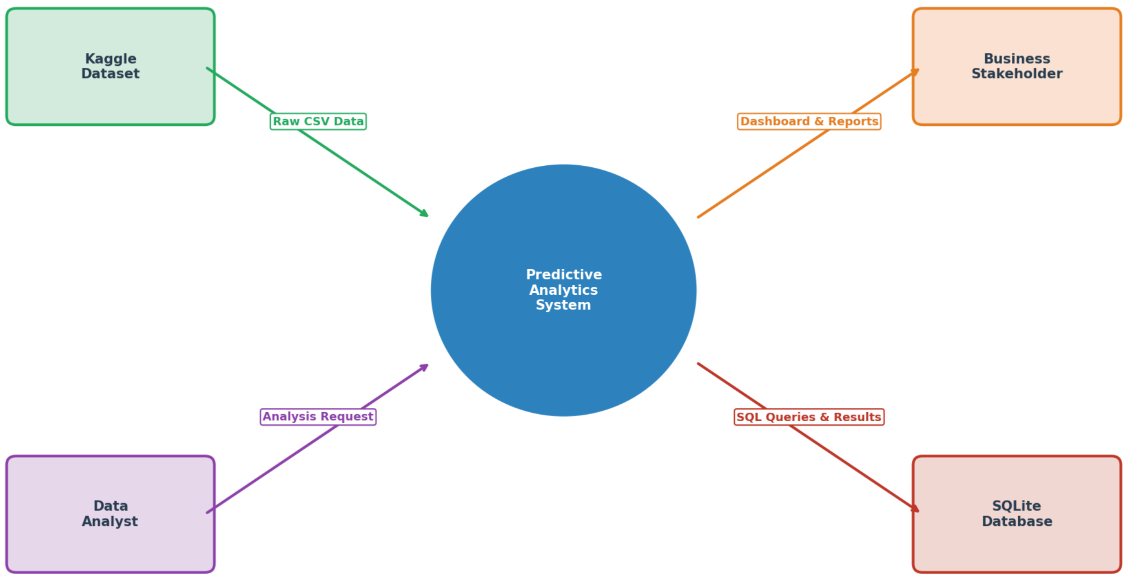


Figure 3.2: DFD Level 0 - Context Diagram

DFD Level 1

The Level 1 DFD further breaks down the system into its key processes:

1. Data Ingestion and Exploration,
2. Data Preprocessing,
3. Model Training and Evaluation,
4. SQL Analysis
5. Dashboard Generation. Data stores (raw data store, processed data store, model output store) and data flows (data transformation) link each process.

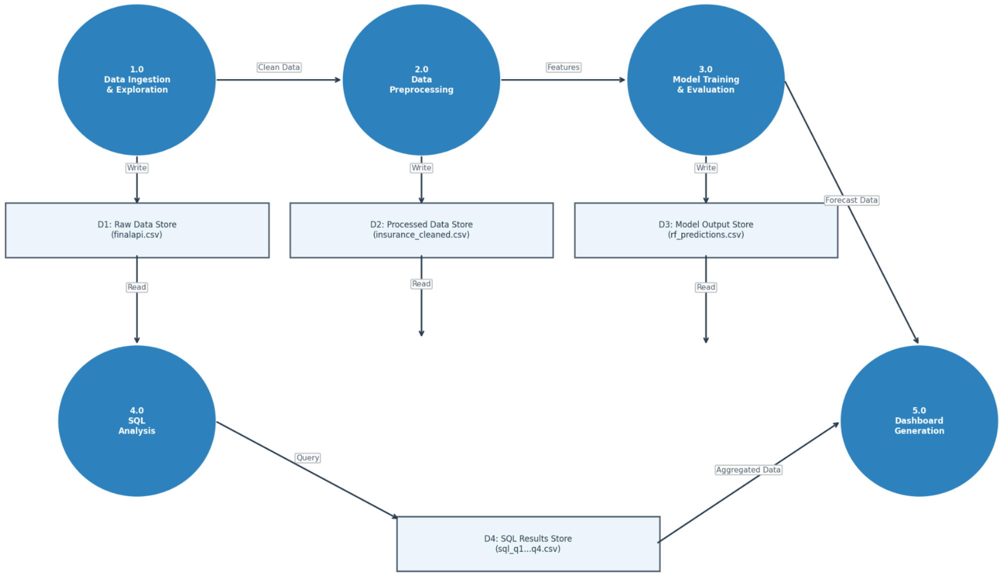


Figure 3.DFD Level 1

3.6 Use Case Diagrams

The key players in the system are Data Analyst and Business Stakeholder. The Data Analyst is the person who communicates with the system in order to load, preprocess, train the model, query the SQL, and create the dashboard. The Business Stakeholder can use the system to view forecasts, filter by year/product line and export reports.

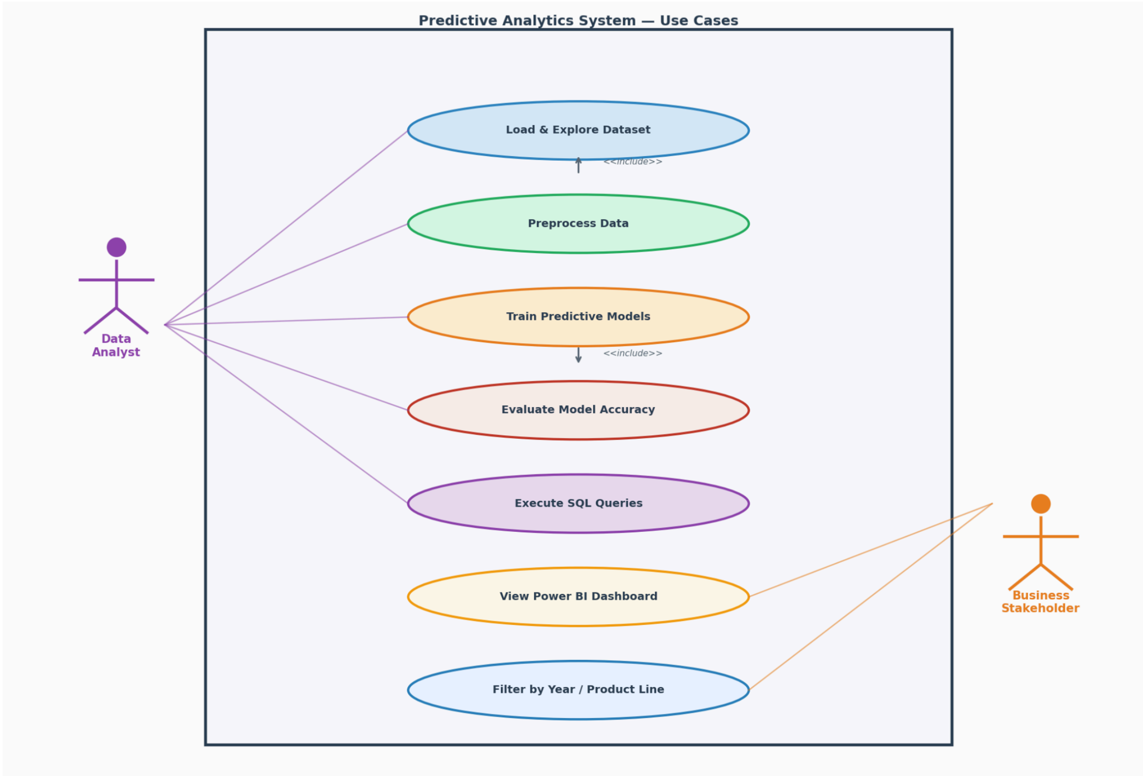


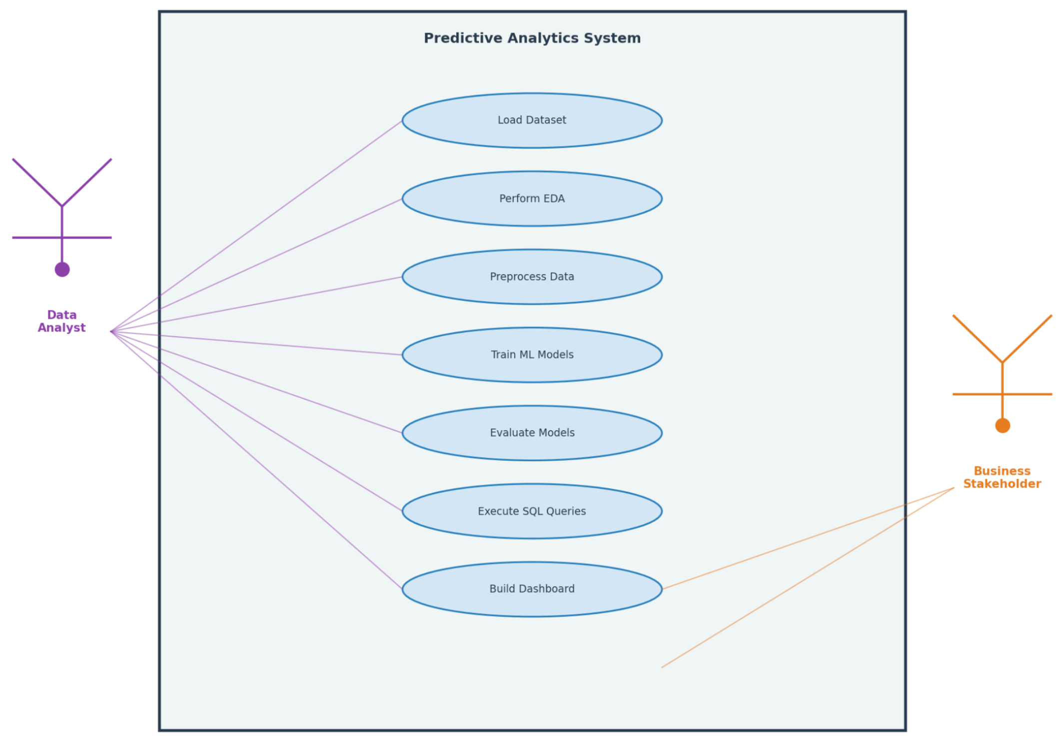
Figure 3.4: Use Case Diagram

CHAPTER 4: SYSTEM DESIGN

4.1 UML Diagrams

4.1.1 Use Case Diagram

The following Use Case Diagram illustrates the interactions between the Data Analyst and the various Use Cases.The following Use Case Diagram shows the interactions between the Data Analyst and the Use Cases: Load Data, Preprocess, Train Model, Evaluate, Query SQL, Build Dashboard.



Use Case Diagram - Actor: Data Analyst, Use Cases: Load Data, Preprocess, Train Model, Evaluate, Query SQL, Build Dashboard

4.1.2 Class Diagram

This class diagram is used to show the main data structures and their relationships in this project. The main classes are: DataLoader for loading CSV and database, DataPreprocessor for cleaning, encoding and feature engineering, ModelTrainer for encapsulating training logic over the three models, ModelEvaluator for computing MAE, RMSE and R², and DashboardExporter for CSV exporting for Power BI.

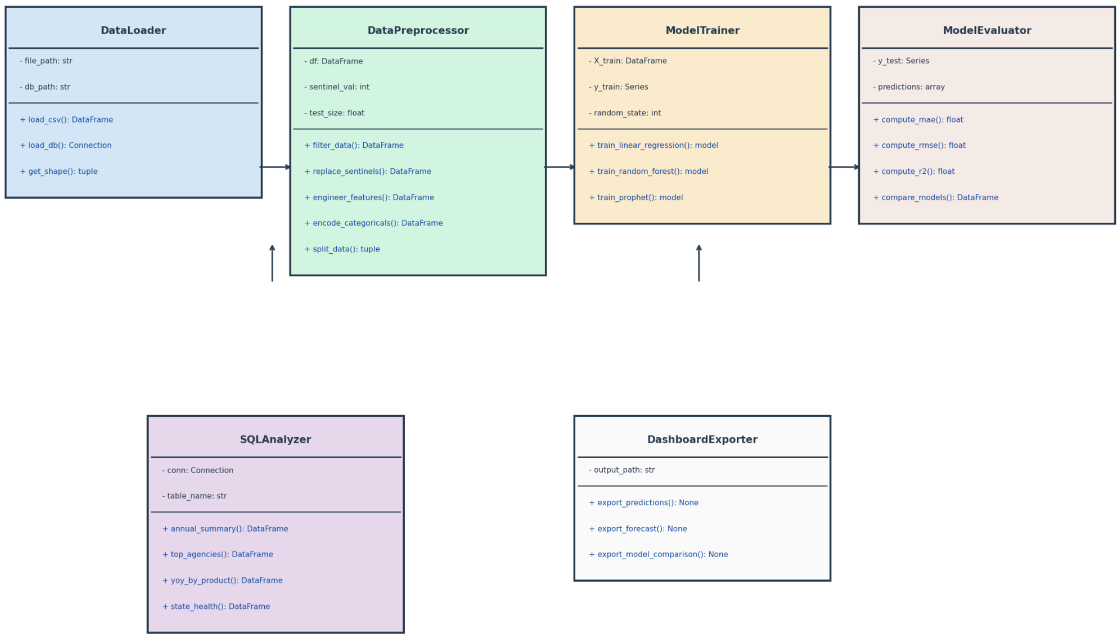


Figure 4.2: Class Diagram

4.1.3 Sequence Diagram

The sequence diagram depicts the interactions between the Data Analyst, Python notebooks, and the database in the model training workflow. The pipeline starts by loading the CSV, passing it to the pre-processing module, continuing with training, evaluation, and finally the prediction module is passed to the CSV for the consumption of

Power BI.

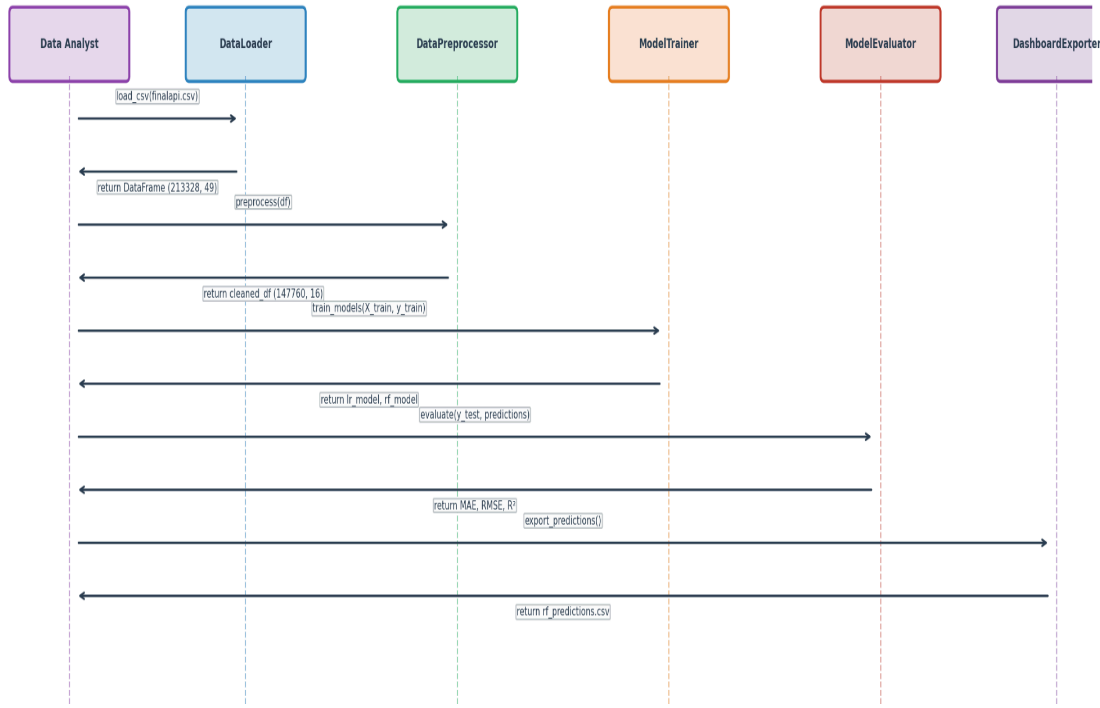


Figure 4.3: Sequence Diagram

4.1.4 Activity Diagram

The activity diagram depicts the entire process, from data loading to the generation of a report, and also the handling of missing data and the selection of the optimal model with the maximum R^2 value.

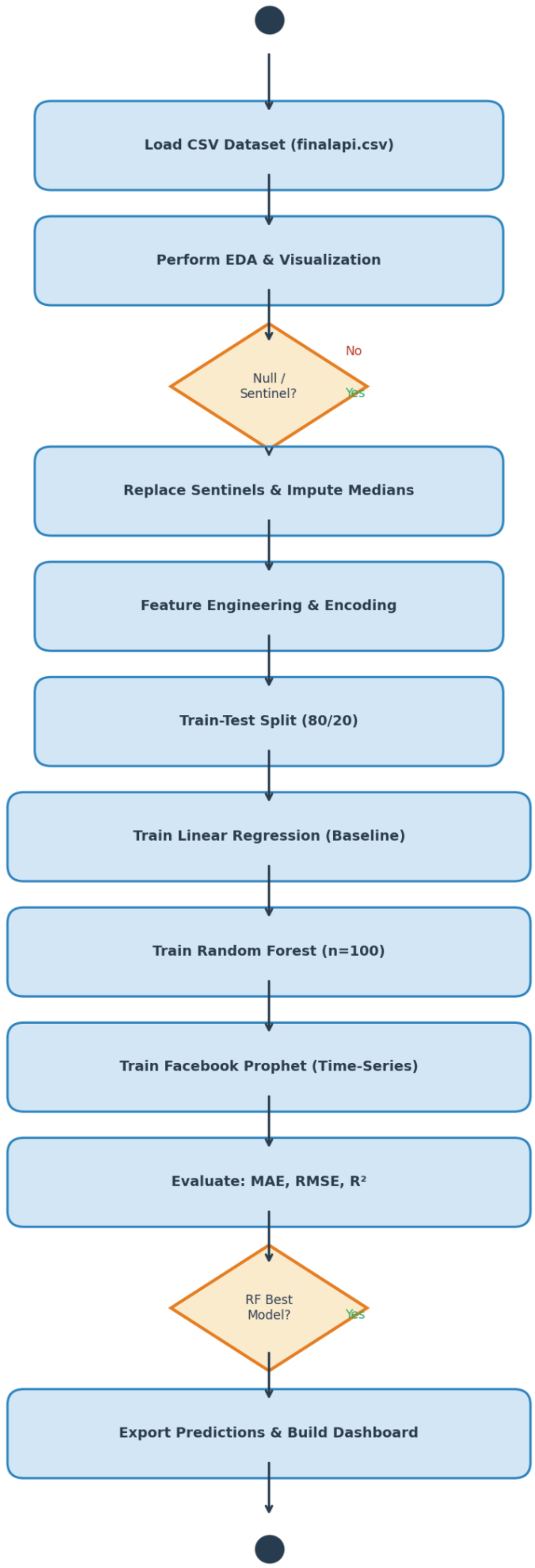


Figure 4.4: Activity Diagram

4.2 ER Diagram

The Entity-Relationship Diagram represents the structure of the insurance_sales table in the SQLite database. The main table is insurance_sales which contains the attributes AGENCY_ID (which is the primary key), STAT_PROFILE_DATE_YEAR, PROD_LINE_ENC, STATE_ENC, WRTN_PREM_AMT, LOSS_RATIO, and RETENTION_RATIO, and some other financial attributes.

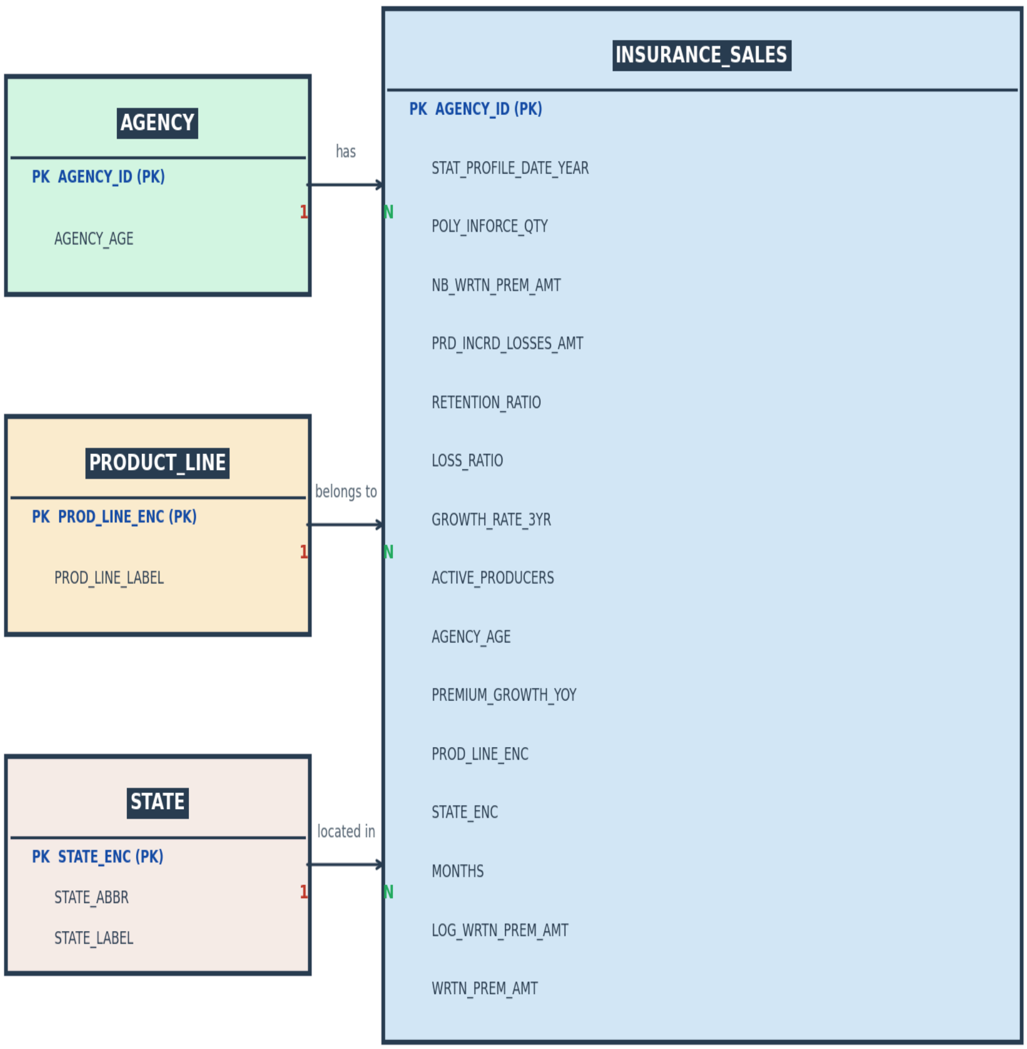


Figure 4.5: ER Diagram

4.3 Database Schema

Column Name	Data Type	Description
AGENCY_ID	INTEGER	Unique identifier for each insurance agency
STAT_PROFILE_DATE_YEAR	INTEGER	Year of the record (2005–2014)
POLY_INFORCE_QTY	INTEGER	Number of policies in force
NB_WRTN_PREM_AMT	REAL	New business written premium amount (USD)
PRD_INCRD_LOSSES_AMT	REAL	Period incurred losses amount (USD)
RETENTION_RATIO	REAL	Customer retention ratio (0–1)
LOSS_RATIO	REAL	Loss ratio (losses/premium)
GROWTH_RATE_3YR	REAL	3-year premium growth rate
ACTIVE_PRODUCERS	INTEGER	Number of active sales producers
AGENCY_AGE	INTEGER	Age of agency (engineered feature)
PREMIUM_GROWTH_YOY	REAL	Year-over-year premium growth (engineered)
PROD_LINE_ENC	INTEGER	Product line encoded (0=CL, 1=PL)
STATE_ENC	INTEGER	State encoded (0–5)
MONTHS	INTEGER	Number of months covered in record
LOG_WRTN_PREM_AMT	REAL	Log-transformed written premium (target)
WRTN_PREM_AMT	REAL	Written premium amount in USD (raw target)

CHAPTER 5: SYSTEM IMPLEMENTATION

5.1 Development Environment

- Operating System: Windows 11
- Programming Language: Python 3.13
- IDE: Jupyter Notebook
- Version Control: Git 2.x with GitHub
- Database: SQLite (via Python sqlite3 module)
- Dashboard: Microsoft Power BI Desktop
- Hardware: Intel Core i5 processor, 8GB+ RAM, 512GB SSD

5.2 Tools and Technologies

Library/Tool	Version	Purpose
pandas	2.x	Data loading, manipulation, aggregation
numpy	1.x	Numerical operations, array processing
matplotlib	3.x	Static chart generation
seaborn	0.x	Statistical visualization (heatmap)
scikit-learn	1.x	Linear Regression, Random Forest, metrics
prophet	1.x	Time-series forecasting
sqlite3	Built-in	SQL database operations
Power BI Desktop	Latest	Interactive dashboard

5.3 Dataset Description and EDA

In this project, I'll be using the Insurance Agency Performance Dataset from Kaggle (moneystore/agencyperformance). It includes performance data for 1,623 independent insurance agencies, a group of 10 states with insurance companies that provide property, casualty, life and brokerage products in Ohio (OH), Pennsylvania (PA), Kentucky (KY), Indiana (IN), West Virginia (WV) and Michigan (MI).

Attribute	Value
Total Records	213,328
Total Features	49
Time Period	2005–2015
Unique Agencies	1,623
States Covered	6 (OH, PA, KY, IN, WV, MI)
Product Lines	Commercial (CL) and Personal (PL)
Null Values	Zero (complete dataset)
Duplicate Records	Zero

5.3.1 Annual Premium Trend Analysis

The total written premium for 2005-2014 is illustrated in Figure 5.1. The overall growth of the period is found from 2005 ($274.8M) to 2006 ($413.4M) with an annual growth of 50.5%. This was followed by a stable plateau from 2006 to 2014 during which total premiums were consistently in the $400-$440M range. The portfolio was largest in 2014 at $437.5M, signifying that the company has a mature and stable insurance portfolio and has a stable customer base.

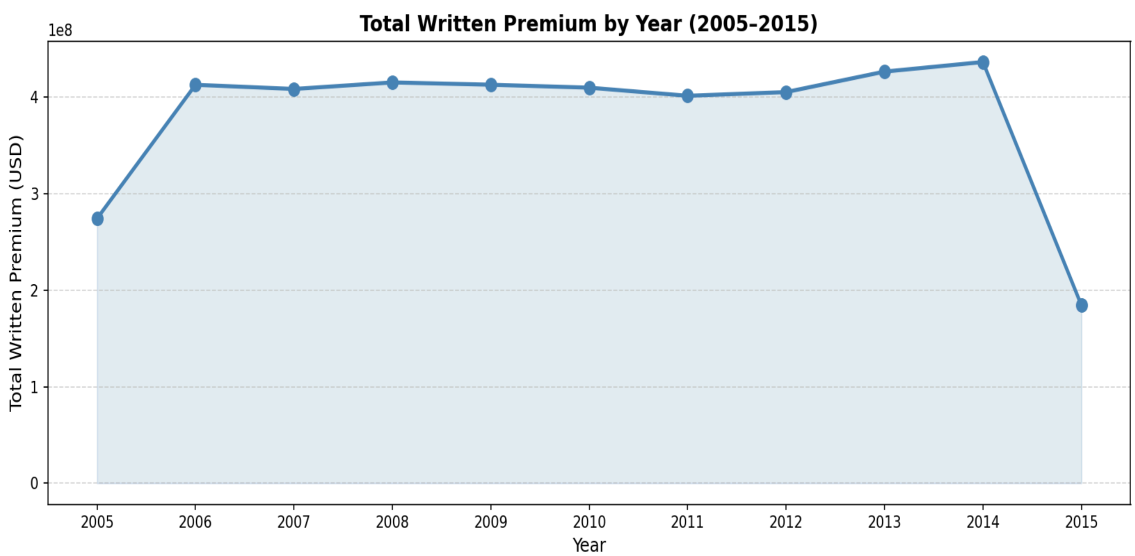


Figure 5.1: Annual Premium Trend

5.3.2 Premium by Product Line

The disparity in trends of Commercial Lines (CL) and Personal Lines (PL) can be seen in the figure 5.2 below. Throughout the period, Personal Lines premium volume consistently was higher than Commercial Lines premium volume. A significant change in strategy is being seen however – Commercial Lines has increased from $95.5M in 2005 to $204.4M in 2014 with 114% growth, while Personal Lines has decreased from $179.4M in 2005 to $233.1M in 2014, with a downtrend after 2006. This indicates a slow transition towards a more commercial insurance product strategy.

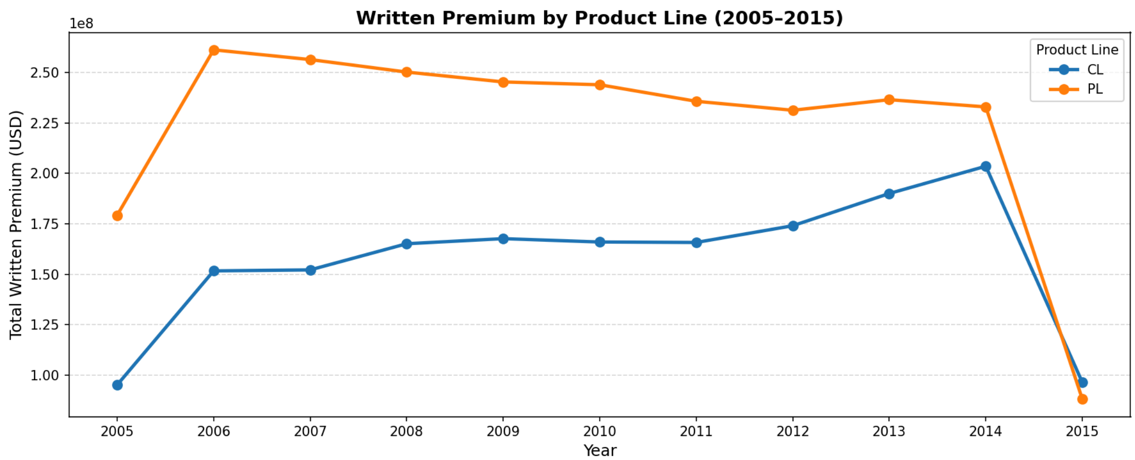


Figure 5.2: Premium by Product Line

5.3.3 Geographic Premium Distribution

The total written premiums by state for the entire study period are shown in Figure 5.3. The Ohio (OH) state portfolio makes up about 58% of the total written premium, led by Ohio. Pennsylvania ($554.9M), Kentucky ($491.8M) and Indiana ($454.7M) are next with significantly lower volumes. Michigan is the lowest contributor with $45.7M. This geographical focus implies to the portfolio risk and indicates the expansion strategy needed in the lower penetration states.

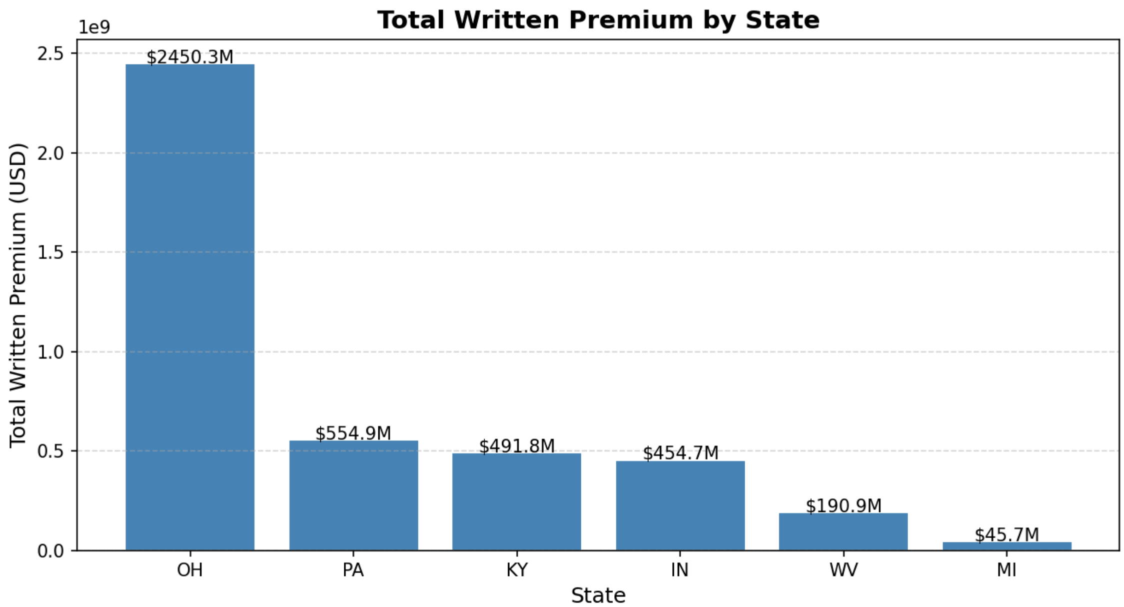


Figure 5.3: Premium by State

5.3.4 Loss Ratio and Retention Analysis

This graph (Figure 5.4) shows the average loss ratio and the average retention ratio over time, plotted on a dual axis. Average retention ratio for the portfolio is 88.75%, a sign of high levels of customer affinity. However, the loss ratio is very volatile, with an unusual high peak in 2012 which could be a sign of increased claims period. Loss ratios were pushed back to more normalized levels in 2013-2014 following the increase in 2012 which indicated that underwriting corrections were successful.

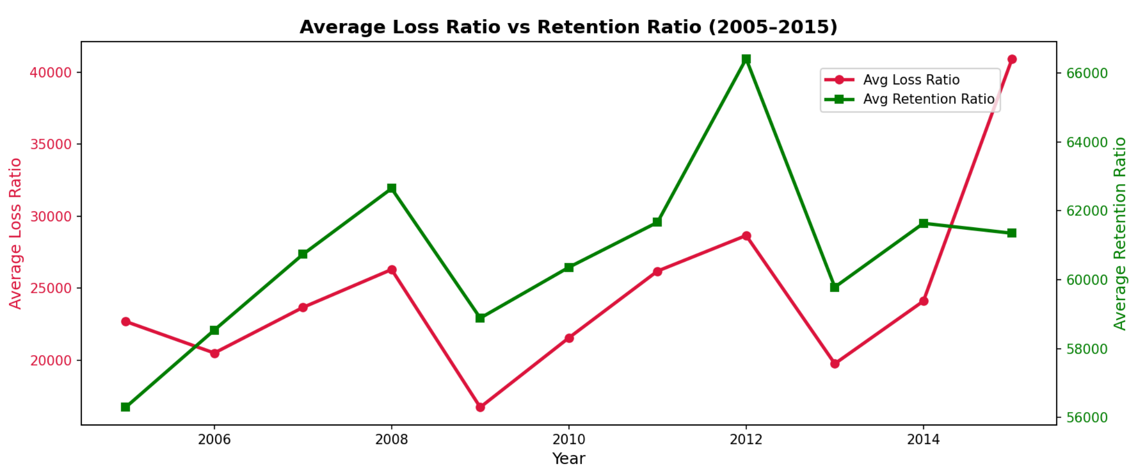


Figure 5.4: Loss vs retention

5.3.5 Premium Distribution Analysis

As shown in figure 5.5, the written premium distribution is very skewed towards the right. The raw distribution reveals that most of the records have very little premium value (close to $0), and high-performing agencies earn premiums of more than $1M. The distribution of the target variable is close to normal after log-transforming, confirming the use of log1p transformation prior to model training. This transformation greatly enhanced the performance of the models.

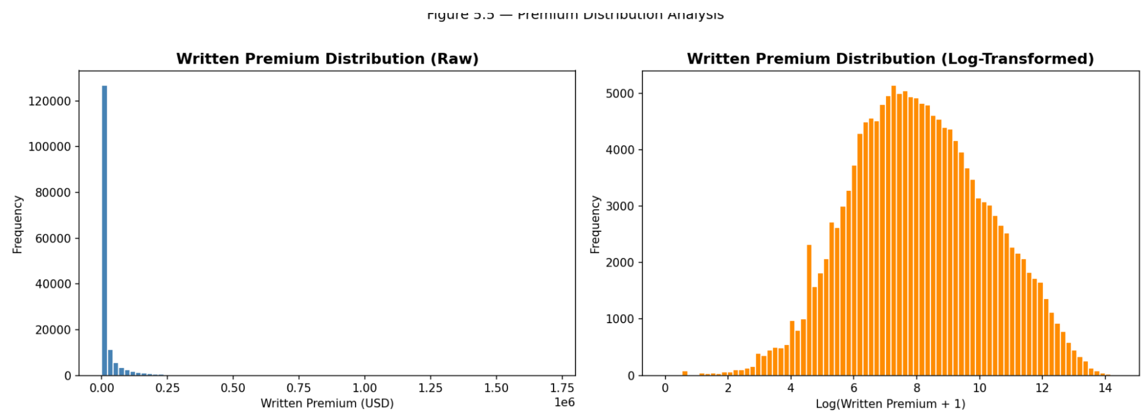


Figure 5.5: Premium Distribution

5.3.6 Correlation Analysis

The correlation heatmap for key financial indicators is given in Figure 5.6. The most important result is that there is almost perfect correlation (r = 1.00) between PRD_ERND_PREM_AMT and WRTN_PREM_AMT, data leakage (both measure the same underlying construct). The column was not included in the features. The number of active policies (POLY_INFORCE_QTY) has the most strong positive correlation with the target variable (r = 0.81), indicating that the most significant predictor of premium volume is the number of active policies.

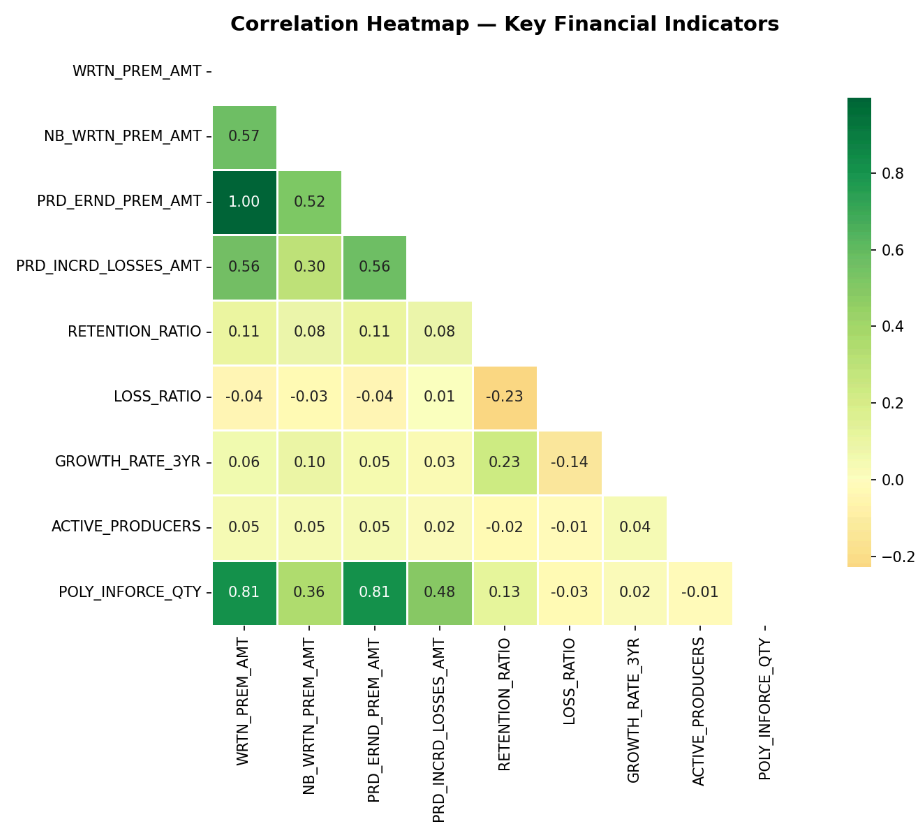


Figure 5.6: Correlation Heatmap

5.4 Data Preprocessing

5.4.1 Data Filtering